Session-09: Exponential Distribution

1. Suppose that the length of a phone call in minutes is an exponential random variable with mean = 10. Find the probability that one has to wait

(i) more than 10 minutes. Ans:0.3688
(ii) between 10 and 20 minutes. Ans: 0.2325

2. The kilometres run(in 1000 of kms) without any sort of problems in respect to certain vehicle is a random variable having pdf $f(x) = \begin{cases} \frac{1}{40}e^{-\frac{x}{40}} & x \geq 0 \\ 0 & x < 0 \end{cases}$.

Find the probability that

(i) At least for 25000 kms. Ans: 0.5353
(ii) At most for 25000 kms. Ans: 0.4647
(iii) Between 16000 to 32000 kms. Ans: 0.2209

3. In a distributed system the time until failure of a node follows an exponential distribution with a mean time to failure of 500 hours.

i. Calculate the probability that a node will fail within the first 100 hours.
Ans: 0.1812
ii. Determine the probability that a node will operate without failure for more than 600 hours. Ans: 0.6989